PlantCare AI

Updated Source Code and Expanded Project Explanation

College AI Immersion Project | Responsive plant-leaf image preview and plant-care guidance prototype

What has been updated

This updated version improves the original prototype with clearer image validation, a reset option, a more transparent demo-result message, better dashboard counting, and expanded explanations. It remains a front-end demonstration: it does not contain a trained AI model and does not diagnose the uploaded leaf image.

Project Explanation

1. Project Overview

PlantCare AI is a responsive website prototype that demonstrates how a digital plant-care interface could work. A user can select a plant-leaf image, preview it, and view general educational guidance. The sample result is fixed demonstration content and is not generated from the uploaded image.

2. Problem Statement

Plant diseases and other plant-health problems can reduce crop quality and yield. Manually inspecting many plants takes time, and visible symptoms may look similar even when their causes differ. A digital interface can help organize image uploads, observations, and care information, while a real diagnosis would require a properly trained and validated model or expert assessment.

3. Proposed Solution

The website includes Home, AI Detection, Care Steps, Prevention, and About sections. Users can select an image from their device, preview it, run the demonstration, read sample tomato early-blight care guidance, and reset the form for another image. The interface works on desktop and mobile screens.

4. Main Features

• Responsive green agriculture-themed layout for mobile and desktop.

• Section navigation without reloading the page.

• Plant-leaf image selection and preview.

• Image-type validation and a clear message when no image is selected.

• Demonstration result with explicit warning that it is fixed sample data.

• General plant-care and prevention information.

• Dashboard counters for demonstration runs and sample results.

• Reset option to clear the selected image and result.

5. Technologies Used

HTML defines the content and page sections. **CSS** controls the colors, layout, cards, spacing, and responsive design. **JavaScript** handles navigation, file validation, image preview, the demo result, reset behavior, and

dashboard counters.

6. How to Run

• Copy the complete code below into a file named index.html.

• Open the file in a modern web browser.

• Choose AI Detection and select a plant-leaf image.

• Check that the image preview appears.

• Press Run Demo Result to display the fixed sample result.

• Open Care Steps or Prevention to read the educational information.

• Press Reset to clear the selected image and begin again.

7. Expected Outcome

The user can navigate the site, choose and preview an image, view a clearly labelled sample result, read care suggestions, and see the demo counter update. The prototype is useful for demonstrating the planned workflow, not for making real crop-treatment decisions.

8. Limitations

No trained model or backend is included. The sample result “Tomato — Early Blight” is fixed and is shown regardless of the uploaded image. No real confidence score is calculated. The image is only previewed in the browser.

9. Future Enhancements

• Train and evaluate a plant-disease classifier using a suitable, labelled image dataset.

• Connect the model through a backend API and return predictions based on image content.

• Validate model performance using separate test data and report appropriate metrics.

• Add crop-specific guidance reviewed by agriculture professionals.

• Support regional languages and accessible voice features.

• Add secure scan history only with appropriate privacy protections.

10. Safety Note

This project is for educational demonstration only. Plant symptoms can have different causes. Confirm serious, unfamiliar, or spreading problems with a qualified agriculture professional before deciding on treatment.

Complete Updated Source Code

Save the following complete code as index.html. It contains the HTML, CSS, and JavaScript in one file, so no extra files are required.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="PlantCare AI student prototype for plant image preview and care guidance.
">
  <title>PlantCare AI | Smart Plant Care</title>
  <style>
    :root {
      --green: #087a45;
      --dark: #063b28;
      --light: #f1fff5;
      --mint: #e8f8ee;
      --text: #17352a;
      --muted: #5c7166;
      --white: #ffffff;
      --border: #cfe8d8;
      --warning: #fff4d5;
    }
    * { box-sizing: border-box; margin: 0; padding: 0; }
    body {
      font-family: Arial, Helvetica, sans-serif;
      background: var(--light);
      color: var(--text);
      line-height: 1.6;
    }
    header {
      background: var(--green);
      color: var(--white);
      padding: 16px;
      text-align: center;
      position: sticky;
      top: 0;
      z-index: 5;
      box-shadow: 0 3px 12px #0002;
    }
    header h1 { font-size: 1.6rem; }
    nav {
      display: flex; flex-wrap: wrap; justify-content: center;
      gap: 8px; margin-top: 12px;
    }
    nav button, .btn {
      border: 0; border-radius: 10px; padding: 10px 14px;
      font-weight: bold; cursor: pointer; font-size: .95rem;
    }
    nav button { background: var(--white); color: var(--green); }
    button:focus-visible, input:focus-visible {
      outline: 3px solid #f0bd4b; outline-offset: 3px;
    }
    .page { display: none; padding: 24px 16px; }
    .page.active { display: block; }
    .container { max-width: 1000px; margin: 0 auto; }
    .hero {
      max-width: 1000px; margin: 0 auto; padding: 46px 22px;
      min-height: 320px; border-radius: 22px; color: white;
      text-align: center; background: linear-gradient(135deg, #07552f, #0b9b59);
      display: flex; flex-direction: column; align-items: center; justify-content: center;
    }
    .hero .leaf { font-size: 3.5rem; }
    .hero h2 { font-size: clamp(2rem, 5vw, 3rem); margin: 8px 0; }
    .hero p { max-width: 720px; margin-top: 10px; }
    .btn { margin: 12px 5px 0; }
    .btn.green { background: var(--green); color: white; }
    .btn.white { background: white; color: var(--green); }
    .btn.secondary { background: #e8eee9; color: var(--text); }
    .section-title { margin: 28px 0 14px; }
    .cards { display: grid; grid-template-columns: repeat(3, minmax(0, 1fr)); gap: 15px; }
    .card, .stat {
      background: white; padding: 20px; border: 1px solid #00000008;
      border-radius: 16px; box-shadow: 0 5px 18px #0000000c; margin: 12px 0;
    }
    .card h3 { margin-bottom: 8px; color: var(--green); }
    .stats h2 { color: var(--green); font-size: 2rem; }
    .stat { text-align: center; }
    .upload {
      border: 2px dashed #58ae7d; padding: 25px; text-align: center;
      border-radius: 16px; background: #f8fffa;
    }
    input[type="file"] { display: block; max-width: 100%; margin: 12px auto; }
    #preview, #resultImage {
```

```
      display: none; max-width: 100%; max-height: 300px; object-fit: contain;
      margin: 18px auto 0; border-radius: 14px;
    }
    .result { display: none; }
    .badge {
      display: inline-block; background: var(--mint); color: var(--green);
      border-radius: 20px; padding: 4px 10px; font-weight: bold; margin: 6px 0 12px;
    }
    .warning { background: var(--warning); padding: 15px; border-radius: 12px; margin-top: 16px; }
    .solution {
      display: flex; gap: 12px; padding: 14px; margin: 10px 0;
      background: #effaf3; border-radius: 12px;
    }
    .number {
      flex: 0 0 32px; width: 32px; height: 32px; display: grid; place-items: center;
      background: var(--green); color: white; border-radius: 50%; font-weight: bold;
    }
    .muted { color: var(--muted); font-size: .93rem; }
    footer { background: var(--dark); color: white; text-align: center; padding: 22px 15px; margin-top: 20px;
}
    @media (max-width: 700px) {
      .cards { grid-template-columns: 1fr; }
      .page { padding: 18px 12px; }
      .hero { padding: 34px 16px; }
      nav button { font-size: .85rem; padding: 8px 10px; }
    }
  </style>
</head>
<body>
  <header>
    <h1>PlantCare AI</h1>
    <nav aria-label="Main navigation">
      <button type="button" onclick="showPage('home')">Home</button>
      <button type="button" onclick="showPage('detect')">AI Detection</button>
      <button type="button" onclick="showPage('solution')">Care Steps</button>
      <button type="button" onclick="showPage('prevention')">Prevention</button>
      <button type="button" onclick="showPage('about')">About</button>
    </nav>
  </header>

  <main>
    <section id="home" class="page active">
      <div class="hero">
        <div class="leaf" aria-hidden="true">Plant Health</div>
        <h2>PlantCare AI</h2>
        <h3>Plant Image Preview and Care Guidance</h3>
        <p>Select a plant-leaf photo, preview it, and explore sample plant-care guidance for a smart-farming
project demonstration.</p>
        <div>
          <button class="btn white" type="button" onclick="showPage('detect')">Upload Leaf Image</button>
          <button class="btn white" type="button" onclick="openCamera()">Take Photo</button>
        </div>
      </div>
      <div class="container">
        <h2 class="section-title">Smart Farming with AI</h2>
        <div class="cards">
          <article class="card"><h3>Image Selection</h3><p>Select a supported image and see its preview
before continuing.</p></article>
          <article class="card"><h3>Detection Demo</h3><p>View a fixed sample result to demonstrate a
possible future AI workflow.</p></article>
          <article class="card"><h3>Care Guidance</h3><p>Read general prevention ideas and educational tomato-
care steps.</p></article>
        </div>
        <h2 class="section-title">Demo Dashboard</h2>
        <div class="cards stats">
          <div class="stat"><h2 id="total">0</h2><p>Demo Runs</p></div>
          <div class="stat"><h2 id="healthy">0</h2><p>Healthy Results</p></div>
          <div class="stat"><h2 id="diseased">0</h2><p>Sample Disease Results</p></div>
        </div>
      </div>
    </section>

    <section id="detect" class="page">
      <div class="container">
        <h2>Plant Leaf Image Demo</h2>
        <p class="muted">Choose an image or use the camera/file chooser on a supported device.</p>
        <div class="card">
          <div class="upload">
            <h3>Select a Plant Leaf Image</h3>
            <p>Choose a JPG, PNG, or another browser-supported image.</p>
            <input type="file" id="leafInput" accept="image/*" aria-label="Choose a plant leaf image">
            <button class="btn green" type="button" onclick="openCamera()">Take Photo</button>
            <img id="preview" alt="Preview of selected plant leaf">
          </div>
          <p id="fileStatus" class="muted" role="status" aria-live="polite">No image selected.</p>
          <button class="btn green" type="button" onclick="detectDisease()">Run Demo Result</button>
          <button class="btn secondary" type="button" onclick="resetDemo()">Reset</button>
          <div class="warning"><strong>Demo limitation:</strong> This version does not run a trained AI model.
 The disease name and illustrative value are fixed sample data, not a real prediction.</div>
```

```
        </div>
        <div class="card result" id="result" aria-live="polite">
          <h2>Illustrative Sample Result</h2>
          <img id="resultImage" alt="Selected leaf image shown alongside sample result">
          <p><strong>Sample plant:</strong> Tomato</p>
          <p><strong>Sample result:</strong> Early Blight</p>
          <p><strong>Illustrative value:</strong> 94% (fixed demo value; not calculated from the image)</p>
          <span class="badge">Educational sample only</span>
          <p><strong>Example symptoms:</strong> Dark spots on older leaves, yellowing around spots, and
gradual leaf damage.</p>
          <button class="btn green" type="button" onclick="showPage('solution')">View Care Steps</button>
        </div>
      </div>
    </section>

    <section id="solution" class="page">
      <div class="container">
        <h2>Care Steps: Sample Tomato Early Blight</h2>
        <p class="muted">General educational suggestions only. This page does not confirm a diagnosis.</p>
        <div class="card">
          <div class="solution"><span class="number">1</span><div><strong>Inspect the plant.</
strong><p>Observe changes in leaves and stems and keep notes on whether symptoms spread.</p></div></div>
          <div class="solution"><span class="number">2</span><div><strong>Remove affected material carefully.
</strong><p>Follow local agricultural guidance for handling and disposing of affected plant material.</p></
div></div>
          <div class="solution"><span class="number">3</span><div><strong>Support good airflow.</
strong><p>Use suitable plant spacing to reduce prolonged moisture around leaves.</p></div></div>
          <div class="solution"><span class="number">4</span><div><strong>Water appropriately.</
strong><p>Water according to the crop and soil needs, avoiding unnecessary wetting of foliage.</p></div></div>
          <div class="solution"><span class="number">5</span><div><strong>Seek qualified advice.</
strong><p>Ask an agriculture professional before deciding on treatment or using any crop product.</p></div></
div>
        </div>
      </div>
    </section>

    <section id="prevention" class="page">
      <div class="container">
        <h2>Prevention and Plant Care</h2>
        <div class="cards">
          <article class="card"><h3>Regular Inspection</h3><p>Check leaves and stems for unusual spots,
yellowing, wilting, or growth changes.</p></article>
          <article class="card"><h3>Water Management</h3><p>Adjust watering to the crop, soil, weather, and
growth stage.</p></article>
          <article class="card"><h3>Balanced Nutrition</h3><p>Choose fertilizer based on crop requirements
and reliable soil advice.</p></article>
          <article class="card"><h3>Healthy Planting Material</h3><p>Use healthy seeds or seedlings and
follow recommended crop hygiene practices.</p></article>
          <article class="card"><h3>Good Airflow</h3><p>Follow suitable plant spacing and manage weeds where
appropriate.</p></article>
          <article class="card"><h3>Expert Support</h3><p>Ask a qualified agriculture professional if
symptoms spread quickly or the cause is unclear.</p></article>
        </div>
        <div class="warning"><strong>Important:</strong> Confirm serious or unfamiliar plant problems with a
qualified agriculture professional.</div>
      </div>
    </section>

    <section id="about" class="page">
      <div class="container">
        <h2>About PlantCare AI</h2>
        <div class="card"><h3>Problem</h3><p>Manual plant inspection can take time, especially across large
fields, and symptoms may have several possible causes.</p></div>
        <div class="card"><h3>Proposed Solution</h3><p>This prototype demonstrates image selection, preview,
a sample result, and educational care guidance. It does not classify the image using AI.</p></div>
        <div class="card"><h3>Future Scope</h3><p>Integrate a trained image-classification model through a
backend API, validate performance, and add verified crop-specific guidance.</p></div>
        <div class="card"><h3>Technology Used</h3><p>HTML structures the website, CSS styles the responsive
interface, and JavaScript controls navigation and demo interactions.</p></div>
      </div>
    </section>
  </main>

  <footer>
    <strong>PlantCare AI</strong><br>
    Smart Plant Care | College AI Immersion Project
  </footer>

  <script>
    let totalScans = 0;
    let healthyResults = 0;
    let diseaseResults = 0;
    let currentImageUrl = null;

    const leafInput = document.getElementById("leafInput");
    const preview = document.getElementById("preview");
    const resultImage = document.getElementById("resultImage");
    const result = document.getElementById("result");
    const fileStatus = document.getElementById("fileStatus");
```

```
    function showPage(pageId) {
      document.querySelectorAll(".page").forEach(function (page) {
        page.classList.remove("active");
      });
      const selectedPage = document.getElementById(pageId);
      if (selectedPage) selectedPage.classList.add("active");
      window.scrollTo({ top: 0, behavior: "smooth" });
    }

    leafInput.addEventListener("change", function () {
      const file = leafInput.files && leafInput.files[0];
      result.style.display = "none";

      if (currentImageUrl) {
        URL.revokeObjectURL(currentImageUrl);
        currentImageUrl = null;
      }

      if (!file) {
        preview.style.display = "none";
        fileStatus.textContent = "No image selected.";
        return;
      }

      if (!file.type.startsWith("image/")) {
        leafInput.value = "";
        preview.style.display = "none";
        fileStatus.textContent = "Please select a valid image file.";
        alert("Please select a valid image file.");
        return;
      }

      currentImageUrl = URL.createObjectURL(file);
      preview.src = currentImageUrl;
      resultImage.src = currentImageUrl;
      preview.style.display = "block";
      fileStatus.textContent = "Selected image: " + file.name;
    });

    function openCamera() {
      showPage("detect");
      setTimeout(function () {
        leafInput.setAttribute("capture", "environment");
        leafInput.click();
      }, 150);
    }

    function detectDisease() {
      if (!leafInput.files || !leafInput.files[0]) {
        alert("Please upload or take a plant leaf photo first.");
        return;
      }

      totalScans += 1;
      diseaseResults += 1;
      document.getElementById("total").textContent = totalScans;
      document.getElementById("healthy").textContent = healthyResults;
      document.getElementById("diseased").textContent = diseaseResults;

      resultImage.src = currentImageUrl || preview.src;
      result.style.display = "block";
      result.scrollIntoView({ behavior: "smooth", block: "start" });
    }

    function resetDemo() {
      leafInput.value = "";
      preview.removeAttribute("src");
      resultImage.removeAttribute("src");
      preview.style.display = "none";
      result.style.display = "none";
      fileStatus.textContent = "No image selected.";
      if (currentImageUrl) {
        URL.revokeObjectURL(currentImageUrl);
        currentImageUrl = null;
      }
    }
  </script>
</body>
</html>
```

Code Explanation: Important Parts

1. HTML structure: the header contains the navigation buttons. Each section has a unique id, such as home, detect, solution, prevention, and about. JavaScript displays one section at a time.

2. CSS design: CSS variables store the theme colors. Flexbox and CSS Grid arrange buttons and cards. The media query changes the card layout to one column on smaller screens.

3. Image input: the file input accepts image files. JavaScript checks that the selected file is an image, creates a temporary browser URL, and displays the preview.

4. Demo result: the Run Demo Result button checks that an image is selected, updates the dashboard counters, and displays the sample result. It does not analyse the image pixels.

5. Reset feature: Reset clears the selected file, hides the preview and result, and releases the temporary image URL.

6. Dashboard: totalScans counts demo runs. diseaseResults counts the fixed sample disease results. healthyResults remains zero because this prototype does not classify images.

Suggested Project Demonstration

• Open the home page and introduce the PlantCare AI project.

• Go to AI Detection and select a leaf photo.

• Show the image preview and explain that it is displayed in the browser.

• Press Run Demo Result and explain that the result is fixed demonstration data.

• Show the care steps and prevention sections.

• Explain future scope: integrate a trained model and validate it before using it for real diagnosis.

Presentation reminder: clearly state that the current version is a front-end prototype. Do not claim the sample result or 94% value is a real AI prediction.